Calibration-Aware Structural Compression of a Point Cloud Transformer via Recoupling-Ratio-Guided Flow Walking

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Abstract

Calibration-aware deep models are desirable for safety-critical applications, yet it remains unclear whether their learned representations can be structurally compressed without sacrificing calibration quality. We investigate this question by adapting the Latent Flow Transformer (LFT) framework of Wu et al. (2025) to a Point Cloud Transformer (PCT) trained on ModelNet40 with the Focal Loss + Multi-class Difference in Confidence and Accuracy (MDCA) objective of Hebbalaguppe et al. (2022). Starting from a 12-layer PCT, we first identify a compressible sequence of Offset-Attention layers using the Recoupling Ratio criterion. The selected block is then replaced by a Flow-Walking-trained latent velocity network that approximates the transformation performed by the original layers, after which the compressed model is fine-tuned end-to-end using the same calibration-aware objective. We evaluate the proposed compression pipeline in terms of classification accuracy, calibration metrics (ECE and SCE), parameter count, and computational cost, assessing the extent to which the calibration and predictive performance of the original model are preserved after compression.

CCS Concepts

• **Computing methodologies → Neural networks**; **Point-based models**; *Computer vision.*

Keywords

point cloud classification, point cloud transformer, model calibration, latent flow transformer, structural compression, flow walking

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1 Introduction

Transformer-based architectures have substantially advanced point cloud understanding by modeling long-range geometric dependencies through self-attention. Among these, the Point Cloud Transformer (PCT) [3] has emerged as a powerful architecture for 3D object recognition, combining local neighborhood aggregation with Offset-Attention to learn expressive geometric representations while naturally preserving the permutation invariance of point clouds. Despite their impressive recognition performance, transformer-based point cloud models remain computationally demanding due to their deep architectures and repeated attention operations, making deployment on resource-constrained and real-time systems increasingly challenging.

To address this limitation, a wide range of model compression techniques—including pruning, quantization, and knowledge distillation—have been proposed to reduce inference cost. While these approaches often achieve significant reductions in model size, they typically require architecture-specific modifications or introduce noticeable degradation under aggressive compression. Recently, Wu *et al.* [2] proposed the Latent Flow Transformer (LFT), a structural compression framework that replaces a contiguous sequence of transformer layers with a single latent transport module learned through Flow Walking. Rather than approximating each layer independently, LFT learns the overall evolution of latent representations across multiple layers, enabling substantial reductions in model complexity while preserving the transformation performed by the original network.

Independently, recent work on neural network calibration has shown that modern deep classifiers frequently produce over-confident predictions despite achieving high classification accuracy. Train-time calibration methods such as the Multi-class Difference in Confidence and Accuracy (MDCA) loss [1] improve the alignment between predicted confidence and empirical correctness by regularizing the confidence distribution during optimization. Unlike post-hoc calibration approaches, MDCA is fully differentiable, requires no separate calibration dataset, and improves prediction reliability without sacrificing classification performance.

Although latent-flow compression and calibration-aware learning have each demonstrated promising results independently, their interaction has not been explored for transformer-based point cloud classification. In particular, it remains unknown whether a calibration-aware Point Cloud Transformer can be structurally compressed while preserving not only its predictive performance but also the calibration benefits obtained during training. Since LFT relies on accurately modeling the evolution of latent representations, understanding its behavior on a calibration-aware teacher network is an important yet unexplored research question.

Motivated by this observation, we investigate the integration of train-time calibration with latent-flow-based structural compression. We first train a 12-layer Point Cloud Transformer using the Focal Loss + MDCA objective to obtain a well-calibrated teacher network. The teacher's latent representations are then analyzed using the Optimal-Transport-based Recoupling Ratio proposed in LFT to identify a highly compressible sequence of Offset-Attention layers. The selected six-layer block is replaced by a lightweight

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Flow-Walking velocity network that approximates the latent transformation performed by the original layers. Finally, the compressed model is fine-tuned end-to-end under the same calibration-aware objective to recover discriminative performance while preserving the computational benefits of structural compression.

Extensive experiments on the ModelNet40 benchmark demonstrate the effectiveness of the proposed framework. The compressed LFT-PCT reduces the number of trainable parameters by **31.0%** and inference FLOPs by **18.5%** relative to the original PCT while recovering classification performance comparable to the calibrated teacher after end-to-end fine-tuning. Furthermore, we evaluate the impact of structural compression on both predictive accuracy and calibration quality using standard calibration metrics, providing the first empirical study of calibration-aware latent-flow compression for transformer-based point cloud classification.

The primary contributions of this work are summarized as follows:

- We present, to the best of our knowledge, the first study that combines train-time calibration with latent-flow-based structural compression for transformer-based point cloud classification.
- We investigate whether a calibration-aware Point Cloud Transformer trained using Focal Loss and MDCA can serve as an effective teacher for Recoupling-Ratio-guided latent-flow compression.
- We demonstrate that replacing six Offset-Attention layers with a single Flow-Walking transport module reduces trainable parameters by **31.0%** and inference FLOPs by **18.5%**, while recovering classification performance after end-to-end fine-tuning.
- We provide an empirical evaluation of the effect of structural compression on both classification performance and calibration, analyzing accuracy together with calibration metrics throughout the compression pipeline.

2 Background and Preliminaries

2.1 Calibration Metrics and the MDCA Loss

Modern deep neural networks often achieve high classification accuracy while producing poorly calibrated confidence estimates, frequently assigning overly confident probabilities to incorrect predictions. Calibration seeks to align a model's predicted confidence with its empirical correctness, thereby improving the reliability of confidence scores without necessarily affecting classification accuracy. Formally, consider a K-class classifier producing a confidence vector $\mathbf{s} \in \mathbb{R}^K$ for an input sample, with predicted class $\hat{y} = \arg\max_j \mathbf{s}[j]$. A classifier is said to be *perfectly calibrated* [1] if, for every confidence value $s \in [0, 1]$,

$$\mathbb{P}(y = y^* \mid \mathbf{s}[y] = s) = s, \tag{1}$$

where y^* denotes the ground-truth class label. Intuitively, Eq. (1) states that predictions assigned a confidence of s should be correct approximately $100s\%$ of the time.

A widely adopted metric for evaluating calibration is the *Expected Calibration Error (ECE)*, which approximates Eq. (1) by partitioning the confidence interval $[0, 1]$ into M equally spaced bins $B_1, B_2, \ldots, B_M$. For each bin, the empirical accuracy A_i is compared against the average predicted confidence C_i, yielding

$$\mathrm{ECE} = \sum_{i=1}^{M} \frac{|B_i|}{N} |A_i - C_i|, \tag{2}$$

where $|B_i|$ denotes the number of samples in the i^{th} confidence bin and N is the total number of evaluation samples. Although ECE is simple and widely used, it evaluates only the confidence assigned to the predicted class and ignores the remaining entries of the confidence vector.

To address this limitation, the *Static Calibration Error (SCE)* extends ECE by measuring calibration independently for every class. Let $A_{i,j}$ denote the empirical accuracy for class j within confidence bin i, and let $C_{i,j}$ represent the corresponding average predicted confidence. SCE is defined as

$$\mathrm{SCE} = \frac{1}{K} \sum_{i=1}^{M} \sum_{j=1}^{K} \frac{|B_{i,j}|}{N} |A_{i,j} - C_{i,j}|, \tag{3}$$

where $|B_{i,j}|$ denotes the number of samples assigned to class j whose confidence lies within the i^{th} bin. Unlike ECE, SCE evaluates the calibration of the complete confidence distribution, making it a more comprehensive metric for multi-class classification.

Despite their usefulness as evaluation metrics, both ECE and SCE rely on confidence binning and absolute-value operations, rendering them non-differentiable and therefore unsuitable as optimization objectives during training. To overcome this limitation, Hebbalaguppe *et al.* [1] proposed the *Multi-class Difference between Confidence and Accuracy (MDCA)* loss, a differentiable regularization term that directly aligns predicted confidence with empirical class frequencies at the mini-batch level.

For a mini-batch containing N_b samples, the MDCA loss is defined as

$$\mathcal{L}_{\mathrm{MDCA}} = \frac{1}{K} \sum_{j=1}^{K} \left| \frac{1}{N_b} \sum_{i=1}^{N_b} s_i[j] - \frac{1}{N_b} \sum_{i=1}^{N_b} q_i[j] \right|, \tag{4}$$

where $s_i[j]$ denotes the predicted confidence assigned to class j for sample i, while

$$q_i[j] = \begin{cases} 1, & \text{if sample } i \text{ belongs to class } j, \\ 0, & \text{otherwise.} \end{cases}$$

Unlike conventional sample-wise ℓ_1 losses, the absolute-value operation in Eq. (4) is applied only after averaging over the entire mini-batch for each class. Consequently, MDCA regularizes the global confidence distribution instead of individual predictions, encouraging the average predicted confidence for every class to match its empirical occurrence within the batch.

The overall optimization objective combines the primary classification loss $\mathcal{L}_C$ with the MDCA regularization term,

$$\mathcal{L}_{\text{total}} = \mathcal{L}_C + \beta \mathcal{L}_{\mathrm{MDCA}}, \tag{5}$$

where β controls the strength of calibration regularization. Throughout this work, $\mathcal{L}_C$ is chosen as Focal Loss with focusing parameter $\gamma = 2$, while $\beta = 10$ following the recommendations of [1]. The resulting calibration-aware Point Cloud Transformer serves as the teacher network for the subsequent latent transport and structural compression stages described in Section 4.

2.2 Point Cloud Transformer

Unlike images, point clouds are unordered sets of irregularly sampled 3D points without an underlying grid structure, making conventional convolutional operations difficult to apply directly. Transformer-based architectures overcome this limitation by modeling long-range dependencies through self-attention while preserving permutation invariance. Among these, the Point Cloud Transformer (PCT) [3] has demonstrated strong performance by combining local neighborhood aggregation with Offset-Attention, enabling both local geometric feature extraction and global contextual reasoning.

Given an input point cloud $P = \{p_1, p_2, \ldots, p_N\} \subset \mathbb{R}^3$, PCT first projects the raw point coordinates into a higher-dimensional feature space through two cascaded Linear–BatchNorm–ReLU (LBR) layers. Since point coordinates alone do not explicitly encode local geometric relationships, feature embedding is followed by two Sampling-and-Grouping (SG) modules that aggregate information from local neighborhoods.

For each sampled anchor point p, the SG module identifies its k-nearest neighbors and constructs a local feature representation by concatenating relative feature offsets with the repeated feature of the anchor point. Formally,

$$\Delta F(p) = \mathrm{concat}_{q \in \mathrm{knn}(p,P)} \left(F(q) - F(p) \right), \tag{6}$$

$$\tilde{F}(p) = \mathrm{concat} \left(\Delta F(p), \mathrm{RP}(F(p), k) \right), \tag{7}$$

$$F_s(p) = \mathrm{MaxPool} \left(\mathrm{LBR}(\mathrm{LBR}(\tilde{F}(p))) \right), \tag{8}$$

where $F(\cdot)$ denotes the point feature representation, $\mathrm{RP}(\cdot, k)$ repeats the anchor feature k times, and MaxPool aggregates neighborhood information into a single local descriptor. This procedure allows PCT to capture fine-grained local geometry while maintaining invariance to the ordering of input points.

Following local feature aggregation, PCT models long-range interactions through a sequence of Offset-Attention (OA) blocks. For an input feature matrix F_{in}, query, key, and value representations are obtained using learned linear projections,

$$(Q, K, V) = F_{\mathrm{in}}(W_q, W_k, W_v), \qquad \tilde{A} = QK^\top, \tag{9}$$

where W_q, W_k, and W_v denote the learnable projection matrices. Unlike conventional transformer attention, which applies a row-wise softmax over the attention matrix, Offset-Attention employs a two-stage normalization strategy. First, attention scores are normalized across the query dimension,

$$\bar{\alpha}_{i,j} = \frac{\exp(\tilde{\alpha}_{i,j})}{\sum_k \exp(\tilde{\alpha}_{k,j})}, \tag{10}$$

followed by ℓ_1 normalization across the key dimension,

$$\alpha_{i,j} = \frac{\bar{\alpha}_{i,j}}{\sum_k \bar{\alpha}_{i,k}}, \tag{11}$$

producing the normalized attention matrix $A = [\alpha_{i,j}]$.

Rather than directly using the attended features, Offset-Attention explicitly models the residual difference between the original features and their attention-weighted aggregation. The output of each attention block is therefore computed as

$$F_{\mathrm{out}} = \mathrm{LBR} \left(F_{\mathrm{in}} - AV \right) + F_{\mathrm{in}}, \tag{12}$$

where the residual connection preserves the original feature representation while the Offset-Attention term captures contextual information from neighboring points. As shown in [3], the quantity $F_{\mathrm{in}} - AV$ approximates the graph Laplacian $(I - A)F_{\mathrm{in}}$, allowing the attention mechanism to emphasize informative geometric variations rather than absolute feature values.

An important characteristic of PCT is that the hidden representations evolve progressively through a sequence of residual Offset-Attention blocks. Consequently, consecutive transformer layers exhibit smooth latent transitions that can be interpreted as samples from an underlying continuous transformation. This property makes PCT particularly well suited for latent transport-based structural compression, where multiple Offset-Attention layers are replaced by a single learned transport operator that approximates the overall latent evolution while preserving the surrounding network architecture.

2.3 Latent Flow Transformer: Flow Matching, the Recoupling Ratio, and Flow Walking

The Latent Flow Transformer (LFT) [2] reformulates the forward propagation of a transformer as a continuous dynamical system, viewing the hidden representations as evolving along a latent trajectory rather than through a sequence of independent discrete layers. Under this interpretation, a contiguous block of transformer layers can be replaced by a single learned transport operator that approximates the latent evolution between the block's input and output representations. Instead of reproducing every intermediate layer individually, the transport operator directly models the overall transformation performed by the entire block, enabling substantial structural compression with reduced computational complexity.

Formally, let h_t denote the hidden representation at continuous time t. The latent evolution is modeled as the solution of an ordinary differential equation (ODE),

$$\frac{dh_t}{dt} = u_\theta(h_t, t), \tag{13}$$

where $u_\theta(\cdot)$ denotes a learnable velocity field parameterized by θ. Integrating Eq. (13) transports the hidden representation from one point in the latent space to another, providing a continuous approximation to the sequence of transformer layers.

Directly solving Eq. (13) during training is computationally expensive because it requires repeated numerical integration. Flow Matching [2] circumvents this difficulty by learning the velocity field directly from paired latent representations, eliminating the need for ODE simulation during optimization.

Consider a transformer block beginning at layer m and ending at layer n. Let x_0 denote the latent representation entering layer m and x_1 the latent representation leaving layer n. A latent state at continuous time $t \in [0, 1]$ is obtained through linear interpolation,

$$x_t = (1-t)x_0 + t\,x_1. \tag{14}$$

The desired velocity along this trajectory is simply

$$v_t = x_1 - x_0,$$

and the velocity estimator is trained by minimizing the Flow Matching objective

$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}_t\left[\|u_\theta(x_t,t) - (x_1 - x_0)\|^2\right]. \tag{15}$$

Here, x_0 and x_1 represent the latent representations at the beginning and end of the compressed transformer block, while $u_\theta(x_t, t)$ predicts the instantaneous transport velocity at time t. Optimizing Eq. (15) enables the model to recover the overall latent transformation performed by the original transformer block without explicitly reconstructing its intermediate layers.

Once the velocity field has been learned, latent representations are transported through numerical integration during inference. Following LFT, a second-order Runge–Kutta (RK2) midpoint integrator is employed,

$$x_{t+d} = x_t + d\,u_\theta\left(x_t + \frac{d}{2}u_\theta(x_t,t),\, t + \frac{d}{2}\right), \tag{16}$$

where d denotes the integration step size. Compared with Euler integration, the midpoint method provides a more accurate approximation of the continuous latent trajectory while maintaining computational efficiency.

Although Flow Matching effectively learns local transport dynamics, its performance deteriorates when multiple latent trajectories intersect. Near such intersections, different training samples produce conflicting velocity targets, causing the learned velocity field to average several possible directions and resulting in inaccurate transport. To quantify this phenomenon before training, LFT introduces the *Recoupling Ratio*, an Optimal Transport (OT)-based metric that estimates the compressibility of a transformer block.

Given latent representations $\{h_m^{(i)}\}$ extracted from layer m and their corresponding representations $\{h_n^{(i)}\}$ from layer n, the optimal transport plan is obtained by solving

$$M = \arg\min_{\gamma} \sum_{i,j} \gamma_{i,j}\, d\left(h_m^{(i)}, h_n^{(j)}\right), \tag{17}$$

where $d(\cdot,\cdot)$ denotes the transport cost and γ represents the transport assignment matrix. The resulting Recoupling Ratio is defined as

$$R = 1 - \mathbb{E}\left[\frac{\mathrm{Tr}(M)}{O_M}\right], \tag{18}$$

where O_M denotes the order of the square transport matrix. A small Recoupling Ratio indicates that the original input-output pairing already agrees closely with the OT-optimal correspondence, implying minimal trajectory crossing and consequently a transformer block that is well suited for latent transport-based compression. Conversely, larger values of R indicate increasingly ambiguous latent transport and reduced learnability through Flow Matching.

To further improve transport learning, particularly for deeper transformer blocks, LFT proposes *Flow Walking (FW)*. Rather than supervising only the local velocity prediction at an interpolated latent state, Flow Walking optimizes the cumulative transport obtained after multiple integration steps,

$$\mathcal{L}_{\mathrm{FW}}(k) = \mathbb{E}_{t_1,\ldots,t_{k-1}}\left[\left\|x_0 + \sum_{i=1}^{k}\Delta_{\theta,t_i} - x_1\right\|^2\right], \tag{19}$$

where

$$\Delta_{\theta,t_i} = s_\theta(x_{t_{i-1}}, t_{i-1}, t_i) - x_{t_{i-1}},$$

$t_0 = 0$, $t_k = 1$, and $s_\theta(\cdot)$ denotes one RK2 integration step given by Eq. (16). Since the increments Δ_{θ,t_i} telescope over successive integration steps, Eq. (19) directly minimizes the discrepancy between the final transported latent representation and the target latent x_1. Consequently, the velocity estimator learns the complete latent trajectory instead of merely fitting local velocity vectors, allowing it to better capture nonlinear transport dynamics over deeper transformer blocks.

In this work, these concepts are adapted from language models to the Point Cloud Transformer. Specifically, the latent representations at the input and output of consecutive Offset-Attention layers are treated as paired states for transport learning. The Recoupling Ratio is first used to identify the most compressible Offset-Attention block, after which a single Flow-Walking-based velocity estimator is trained to replace the selected layers. The resulting compressed backbone is finally fine-tuned using the calibration-aware objective introduced in Section 2.1, enabling substantial reductions in both model parameters and inference FLOPs while preserving classification performance.

3 Experimental Pipeline

The proposed framework integrates calibration-aware training with latent transport-based structural compression to obtain a compact Point Cloud Transformer (PCT) while preserving both predictive performance and model calibration. The complete pipeline consists of four sequential stages: (1) calibration-aware teacher construction, (2) Recoupling Ratio-guided block selection, (3) Flow-Walking-based latent transport learning, and (4) end-to-end fine-tuning of the compressed model. Unless otherwise specified, all experiments employ a 12-layer PCT backbone (embedding dimension 128, attention dimension $d_a = 32$, k-NN size 16, and two Sampling-and-Grouping stages reducing the number of points from $1024 \rightarrow 512 \rightarrow 256$), following the architectural design of [3]. The backbone is intentionally extended from the original four Offset-Attention layers to twelve layers in order to provide sufficiently deep intermediate transformer blocks for latent transport-based compression, analogous to the deeper transformer setting considered in [2].

(1) **Stage 1 — Calibration-aware teacher construction.** Two identical 12-layer PCT models are trained on the ModelNet40 dataset under identical experimental settings, differing only in the optimization objective. The baseline teacher is trained using Cross-Entropy Loss with Label Smoothing ($\alpha = 0.2$), whereas the second teacher employs the calibration-aware objective consisting of Focal Loss ($\gamma = 2$) and the MDCA regularization term (Eq. 5) with $\beta = 10$. This stage aims to obtain a well-calibrated teacher whose latent representations are subsequently used for structural

compression. Empirically, the FL+MDCA-trained teacher consistently produced more favorable Recoupling Ratio values, particularly across the middle Offset-Attention blocks, than the baseline PCT. Since lower Recoupling Ratios indicate reduced trajectory crossing and greater suitability for latent transport learning, the calibrated teacher is selected for all subsequent stages of the pipeline.

(2) **Stage 2 — Recoupling Ratio-guided block selection.** Using the calibrated teacher, latent representations are extracted from the input of every Offset-Attention layer over 50 batches of the training set, corresponding to 1,600 point clouds with pooled latent representations of dimension 256×128. The pairwise Recoupling Ratio between candidate transformer blocks is then computed using the Optimal Transport formulation described by Eqs. (17)–(18). Following the selection criterion proposed in [2], the widest contiguous middle block with $R = 0$ is selected as the compression target, as such blocks exhibit minimal trajectory crossing and are therefore most amenable to latent transport-based approximation.

(3) **Stage 3 — Flow-Walking-based latent transport learning.**
After identifying the target block, the calibrated teacher is frozen, and a single DiT-style velocity estimator (Section 2.3) is trained to learn the latent transformation between the input and output representations of the selected Offset-Attention block. The self-attention parameters of the velocity estimator are initialized from the teacher's source layer, while the remaining transport-specific parameters follow the initialization strategy described in [2]. Training is performed using the Flow-Walking objective (Eq. 19) with $k = 3$, and early stopping is applied based on the validation Normalized Mean Squared Error (NMSE). Rather than reproducing each intermediate transformer layer individually, the learned velocity estimator approximates the complete latent evolution performed by the removed Offset-Attention block through continuous transport.

(4) **Stage 4 — End-to-end fine-tuning.**
Following latent transport learning, the trained velocity estimator is inserted into the backbone in place of the selected Offset-Attention layers, while all remaining network parameters are directly transplanted from the calibrated teacher. The resulting compressed network is then fine-tuned end-to-end using the same calibration-aware objective (Eq. 5) employed during teacher training. This final optimization stage adapts the classifier to the compressed latent representations while preserving the parameter and computational savings achieved through structural compression.

Unlike the original LFT formulation developed for language models, Point Cloud Transformers operate on unordered point sets and therefore lack a canonical token ordering. Consequently, Recoupling Ratio analysis is performed on cloud-level mean-pooled latent representations instead of individual token embeddings. This modification preserves the permutation invariance of the backbone while allowing the latent transport framework to be directly applied to point cloud classification.

4 Results

4.1 Calibration Comparison

The first stage of the proposed pipeline evaluates the effectiveness of calibration-aware training for constructing the teacher network used in subsequent latent transport learning. Two identical 12-layer PCT models are trained under identical experimental settings, differing only in their optimization objective: Cross-Entropy with Label Smoothing (CE+LS) and the proposed calibration-aware objective combining Focal Loss with MDCA (Eq. 5). The resulting models are evaluated on the ModelNet40 test set in terms of both classification performance and calibration quality.

Both teacher models achieve comparable classification accuracy, indicating that incorporating the MDCA regularizer does not significantly affect the discriminative capability of the backbone. However, a substantial improvement is observed in confidence calibration. Compared with the CE+LS baseline, the proposed FL+MDCA objective reduces the Expected Calibration Error by approximately 89.4% and the Static Calibration Error by nearly 59%, while maintaining a similar classification accuracy. The quantitative comparison is summarized in Table 1.

Table 1: Performance comparison of teacher models trained under identical experimental settings on ModelNet40. The proposed FL+MDCA objective significantly improves calibration while maintaining comparable classification accuracy.

Method	Acc. (%)	ECE (%)	SCE ($\times 10^{-3}$)
CE+LS	**90.92**	13.82	9.02
FL+MDCA	90.15	**1.46**	**3.69**

The evolution of calibration during training further supports these observations. While the CE+LS baseline exhibits only modest reductions in calibration error, the FL+MDCA objective steadily decreases ECE from approximately 18% during the initial stages of training to below 2% at convergence. This behavior indicates that the MDCA regularizer effectively aligns predicted confidence with the empirical class distribution throughout optimization.

Given its substantially superior calibration performance, the FL+MDCA-trained model is selected as the teacher for the remaining stages of the proposed framework. As demonstrated in the following section, the calibrated teacher also exhibits more favorable Recoupling Ratio characteristics across the middle Offset-Attention blocks, making it a better candidate for latent transport-based structural compression.

4.2 Recoupling Ratio Analysis

Following teacher training, Recoupling Ratio (RR) analysis is performed to identify transformer blocks that are suitable for latent transport-based structural compression. Guided by the criterion introduced in Section 2.3, blocks with lower RR values exhibit reduced trajectory crossing and are therefore more amenable to replacement by a single learned transport operator.

Empirically, the FL+MDCA-trained teacher consistently produced more favorable RR values across the middle Offset-Attention blocks than the baseline PCT teacher. In particular, the calibrated teacher yielded a greater number of zero-RR candidate blocks, indicating latent representations that are better suited for transport-based approximation. Representative RR values obtained from the calibrated teacher are summarized in Table 2.

Table 2: Representative Recoupling Ratio (RR) values obtained from the calibrated FL+MDCA teacher. Lower RR values indicate transformer blocks that are more suitable for latent transport-based structural compression.

Block	Span	RR	Verdict
OA[0]→OA[1]	1	0.000	Compressible
OA[3]→OA[9]	**6**	**0.000**	**Selected**
OA[4]→OA[10]	6	0.000	Compressible
OA[0]→OA[4]	4	0.641	Avoid
OA[0]→OA[11]	11	0.953	Avoid

Among all candidate transformer blocks, OA[3]→OA[9] constitutes the widest contiguous zero-RR region and is therefore selected for compression. Although the OA[0]→OA[1] block also achieves an RR of zero, it spans only a single Offset-Attention layer and therefore offers negligible compression benefit. In contrast, the six-layer OA[3]→OA[9] block provides substantially greater parameter and computational savings while remaining fully compressible. Conversely, blocks spanning the early Offset-Attention layers over longer ranges exhibit considerably larger RR values, indicating increased trajectory crossing and reduced suitability for replacement by a single transport operator. This behavior is consistent with the observations of [2], where intermediate transformer representations are found to be more amenable to latent transport than the earliest layers.

4.3 Flow-Walking Training and Zero-Shot Evaluation

The selected six-layer Offset-Attention block (OA[3]–OA[8]) is replaced by a single Flow-Walking velocity estimator trained using the procedure described in Section 3. The resulting compression performance is summarized in Table 3.

Table 3: Flow-Walking training and zero-shot evaluation of the compressed model.

Quantity	Value
Compressed block	OA[3]→OA[8]
Identity baseline NMSE	0.3519
Velocity model NMSE	**0.0384**
Teacher accuracy	89.63%
Zero-shot accuracy	68.35%
Teacher FLOPs	1.8070 G
Compressed FLOPs	1.4733 G (−18.5%)
Training FLOPs (Stage 3)	71,443 G

The learned velocity estimator converges rapidly, reaching a validation NMSE of 0.0384, which is approximately nine times lower than the identity baseline. This confirms that the selected zero-RR transformer block can be effectively approximated through a single latent transport operator.

Replacing six Offset-Attention layers with a single Flow-Walking module immediately reduces the computational complexity of the network, decreasing inference cost from 1.8070 GFLOPs to 1.4733 GFLOPs, an 18.5% reduction. Importantly, these computational savings are purely architectural and are therefore preserved both before and after fine-tuning.

Although the compressed network achieves only 68.35% classification accuracy under zero-shot evaluation, this result should be interpreted in the context of structural replacement rather than final model performance. At this stage, the newly introduced velocity estimator has been trained only to reproduce the teacher's latent transformation, while the downstream classifier has not yet been jointly optimized for the modified feature distribution. Consequently, the zero-shot model serves primarily as an initialization for the subsequent end-to-end fine-tuning stage rather than as the final compressed classifier.

4.4 End-to-End Fine-Tuning

Following latent transport training, the compressed network is fine-tuned end-to-end using the same FL+MDCA objective employed for teacher training. Table 4 summarizes the performance of the teacher models, the zero-shot compressed model, and the final fine-tuned network.

Table 4: Performance comparison of the baseline, calibration-aware teacher, and compressed LFT-PCT before and after end-to-end fine-tuning.

Metric	CE+LS	FL+MDCA	Zero-shot	Fine-tuned
Accuracy (%)	90.92	90.15	68.35	**90.40**
ECE (%)	13.82	**1.46**	–	5.67
SCE ($\times 10^{-3}$)	9.02	**3.69**	–	3.91
Parameters	2.71M	2.71M	1.87M	1.87M
FLOPs/sample (G)	1.8070	1.8070	1.4733	1.4733

End-to-end fine-tuning successfully recovers the performance loss introduced by structural compression. The compressed model improves from 68.35% zero-shot accuracy to 90.40%, slightly surpassing the FL+MDCA teacher while retaining the compressed architecture. Unlike the zero-shot model, whose classifier has not yet adapted to the modified latent representations, end-to-end optimization jointly updates the velocity estimator and classification head, allowing the network to effectively exploit the learned transport representation.

Importantly, this recovery in classification accuracy is achieved without sacrificing the computational benefits of compression. Since fine-tuning modifies only the model parameters and not the network architecture, the compressed model permanently retains a 31.0% reduction in trainable parameters (2.71M → 1.87M) together with an 18.5% reduction in inference FLOPs (1.8070G → 1.4733G). These savings are therefore preserved during deployment while maintaining classification performance comparable to the original teacher.

Calibration exhibits a modest degradation after fine-tuning. The ECE increases from 1.46% for the calibrated teacher to 5.67% for the fine-tuned model, indicating that joint optimization prioritizes classification performance over confidence calibration. Nevertheless, the compressed model remains substantially better calibrated than the uncalibrated CE+LS baseline, whose ECE is 13.82%. In contrast, the Static Calibration Error remains nearly unchanged (3.69×10^{-3} versus 3.91×10^{-3}), suggesting that the class-wise confidence structure learned by the calibrated teacher is largely preserved throughout compression and subsequent fine-tuning.

The computational overhead of the proposed compression framework is incurred only once during training. Stage 3 requires approximately 71,443 GFLOPs to train the velocity estimator, whereas Stage 4 fine-tuning requires 8,698,502 GFLOPs owing to end-to-end back-propagation through the compressed network. These training costs do not affect deployment efficiency, as inference is performed using the compressed architecture reported in Table 4.

5 Discussion

5.1 Zero-Shot Performance and the Role of Fine-Tuning

Although the proposed compression framework achieves substantial architectural savings immediately after replacing six Offset-Attention layers with a single Flow-Walking module, the zero-shot compressed model exhibits a noticeable reduction in classification accuracy. This behavior is expected, since the velocity estimator is optimized to reproduce the teacher's latent representation rather than directly optimize the downstream classification objective.

The Flow-Walking objective minimizes the reconstruction error between the transported latent representation and the corresponding teacher latent (Eq. 19). The resulting validation NMSE of 0.0384 confirms that the learned transport operator accurately approximates the overall latent transformation. However, latent reconstruction is fundamentally an average reconstruction objective. Consequently, small discrepancies in highly discriminative regions of the latent space may have only a minor effect on the reconstruction loss while producing a considerably larger impact on the final classification decision. In the proposed framework, the Recoupling Ratio is also evaluated using mean-pooled point-cloud representations, which identify blocks that preserve global latent correspondence but do not explicitly guarantee preservation of fine-grained point-level features that are important for object classification.

The zero-shot model therefore serves as an initialization of the compressed architecture rather than its final operating point. Following structural replacement, the downstream classifier receives a distribution of latent features that differs from that produced by the original transformer block, despite representing the same semantic content. End-to-end fine-tuning allows both the velocity estimator and the remaining network parameters to adapt jointly to this modified feature distribution, thereby restoring the discriminative capability of the compressed model. The experimental results demonstrate this behavior, with classification accuracy recovering from 68.35% to 90.40%, matching and slightly exceeding the calibrated teacher.

More importantly, the proposed compression framework should be evaluated in terms of the final compressed model rather than the intermediate zero-shot initialization. Structural compression is performed before fine-tuning, meaning that the reductions in model complexity are architectural properties of the network rather than consequences of optimization. Consequently, both the zero-shot and the fine-tuned models retain the same compressed architecture, consisting of 31.0% fewer parameters and an 18.5% reduction in inference FLOPs relative to the original teacher. Fine-tuning updates only the model weights and does not alter these computational savings.

It is therefore important to distinguish between the one-time optimization cost of fine-tuning and the computational cost incurred during deployment. Although fine-tuning introduces an additional training stage, it is performed only once after structural compression. In contrast, inference on a held-out test set or during real-world deployment is always carried out using the final fine-tuned compressed model, which retains the same reduced architecture as the zero-shot model. Consequently, every inference benefits from the 31.0% reduction in parameters and 18.5% reduction in FLOPs while achieving classification performance comparable to the original teacher. Therefore, the practical comparison is not between the zero-shot compressed model and the original PCT, but between the final fine-tuned compressed model and the teacher network, where the proposed framework provides equivalent predictive performance at a substantially lower computational cost.

An additional observation is that the calibrated FL+MDCA teacher consistently produced more favorable Recoupling Ratio characteristics than the baseline teacher, particularly across the middle Offset-Attention layers. This suggests that calibration encourages more coherent latent representations, making them better suited for latent transport-based approximation. While this observation is empirical, it provides evidence that calibration contributes not only to improved confidence estimation but also to identifying transformer blocks that are more amenable to structural compression.

6 Future Work

The proposed framework demonstrates that confidence calibration and latent transport-based structural compression can be effectively combined within a point cloud transformer while maintaining competitive classification performance and reducing computational complexity. Several directions remain for future investigation.

First, although this work focuses on object classification using ModelNet40, the proposed framework can be extended to more challenging 3D vision tasks such as semantic segmentation, part segmentation, and large-scale scene understanding. Evaluating the interaction between calibration-aware training and latent transport compression in these settings would provide further insight into the generality of the approach.

Second, the relationship between model calibration and latent compressibility warrants deeper investigation. Our experiments suggest that the calibrated FL+MDCA teacher consistently produces more favorable Recoupling Ratio characteristics across the middle Offset-Attention layers, indicating that calibration may encourage latent representations that are inherently easier to approximate using continuous transport. Establishing a theoretical understanding of this relationship could guide the design of compression-aware calibration objectives.

Another promising direction is the development of Recoupling Ratio-aware training strategies. Rather than identifying compressible blocks only after teacher training, future work could directly encourage transformer representations to produce lower Recoupling Ratios during optimization, enabling architectures that are designed for efficient latent transport from the outset.

Future work may also investigate more expressive latent transport models. The current implementation employs a single Flow-Walking velocity estimator for replacing multiple transformer layers. More powerful transport operators, adaptive integration schemes, or hierarchical latent transport mechanisms may further improve zero-shot reconstruction quality while preserving the computational benefits of structural compression.

Finally, the proposed pipeline can be evaluated on larger transformer architectures and real-world deployment scenarios where reductions in parameter count and inference FLOPs translate directly into lower memory usage, faster inference, and improved energy efficiency. Applying the framework to modern large-scale point cloud foundation models represents a natural extension of the present work.

7 Conclusion

This work presented a unified framework that combines confidence calibration with latent transport-based structural compression for point cloud transformers. A 12-layer Point Cloud Transformer was first trained using the FL+MDCA objective to obtain a well-calibrated teacher, after which Recoupling Ratio analysis was employed to identify compressible transformer blocks for replacement using a single Flow-Walking velocity estimator.

Experimental results demonstrate that calibration-aware training substantially improves confidence estimation, reducing the Expected Calibration Error from 13.82% to 1.46% while maintaining comparable classification accuracy. Furthermore, the calibrated teacher produces latent representations that exhibit favorable Recoupling Ratio characteristics, enabling effective structural compression of the middle transformer layers. Although the compressed model experiences a drop in zero-shot classification performance, end-to-end fine-tuning successfully adapts the compressed architecture, recovering classification performance comparable to the teacher's accuracy while preserving the architectural benefits of compression.

The final compressed model achieves classification performance comparable to the original teacher while reducing the model size by 31.0% and inference FLOPs by 18.5%. Although a moderate increase in ECE is observed after fine-tuning, calibration remains substantially better than the uncalibrated baseline, and class-wise calibration is largely preserved.

Overall, the proposed framework demonstrates that confidence calibration and latent transport-based structural compression are complementary rather than competing objectives. Beyond improving confidence estimation, calibration appears to produce latent representations that are more amenable to transport-based approximation, enabling the construction of smaller and computationally efficient point cloud transformers without sacrificing predictive performance. These findings establish a promising direction for jointly designing reliable and efficient transformer architectures for 3D vision applications.

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